Design FMEA — VBF-T7R1NOFORCE-DFMEA-01

Case t7_r1_noforce — 2026-08-25. Qualitative version, based on simulation signals (S/O three-level; RPN = qualitative S x O). Complete FMEA incl. process/supplier: N/A (beyond pure simulation boundary).

Failure mode	Cause	Simulation signal	Sev	Occ	Mitigation (implemented)
Anode plating @4C	anode kinetics/transport	R1 -0.192 V FAIL -> R5 +0.0153 V PASS	High	Low (design)	4 um particles; transport electrolyte; cooling
Thermal runaway (nail)	insufficient heat rejection	R1 triggered 70-72 s (no cooling) -> R5 triggered=false	High	Low w/ cooling	Liquid cooling h=100 mandatory
SEI growth @45C	solvent-diffusion-limited	R4 551.97 nm FAIL -> R5 510.5 nm (+39.5 margin)	Medium	Medium	thin negative 85.2 um; ALD coating
Electrolyte oxidative decomp.	HOMO vs window	no molecular HOMO value (start_stage=3, real_compute=false)	Medium	Low (qual.)	4.2 V cut-off; DFT follow-up listed
Insufficient capacity	low loading	5.0065 Ah vs 5.0 nominal	Medium	Low	baseline loading
4C thermal overshoot	rate heat vs cooling	T_max 325.54 K vs 573 K	High	Low	h=100 cooling